

Roll No.:.....

ODD END SEMESTER EXAMINATION 2025

Name of the Program: B. Tech (ECE) (ESR and IoT)

Semester: III

Name of the Course: Fundamentals of Computer Organization Applications

Course Code: **TEC-359**

Time:3 Hours

Maximum Marks:100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b &c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1		(20Marks)	
(a)	Explain the structure of a computer system with a neat block diagram. Describe the role of each functional unit.		CO1
(b)	Differentiate between Von Neumann and Non-von Neumann architectures with examples.		CO1
(c)	Explain Moore's law and its significance in the advancement of computer technology.		CO1
Q2			
(a)	Explain Booth's multiplication algorithm with an example.		
(b)	Describe restoring division algorithms with an example.		CO2
(c)	Explain the IEEE 754 format for 64 bits and arrange the number $(65.125)_{10}$ in the mentioned IEEE 754 format.		CO2 CO2 CO2
Q3			
(a)	Write the program to evaluate the expression $X = A [B+C(D+E)] / F(G+H)$ using the zero address instructions and one address instructions.		CO3
(b)	Compare CISC and RISC architectures based on instruction format, complexity, and performance.		CO3
(c)	Discuss different types of operands and their roles in instruction execution.		CO3
Q4			
(a)	Explain Memory Hierarchy Design and its Characteristics according to various parameter.		CO4
(b)	Explain the classification of semiconductor memories. Discuss SRAM and DRAM cell design.		CO4
(c)	Explain average access time of CPU in memory hierarchy and also explain Cache memory.		CO4
Q5			
(a)	Describe the function of registers, ALU, and control unit in a CPU.		CO5
(b)	Explain the hardwired and microprogrammed control units.		CO5
(c)	Discuss the role of control signals and timing in processor operation		CO6

